

WIZARD CHESS

Design and Implementation of an Automated Magnetic Chessboard

Technical Report

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Project theme: concealed mechatronic actuation for automatic chess-piece movement

Abstract

Wizard Chess is a mechatronic chessboard designed to move physical chess pieces automatically while preserving the appearance and interaction of a conventional board. The system uses a concealed two-axis Cartesian gantry beneath a 60 × 60 cm playing surface. A movable electromagnet couples through a 3 mm birch plywood top to permanent magnets embedded in 3D-printed chess pieces. Belt-driven stepper motion positions the electromagnet below a selected piece, the magnet is energized, and the carriage translates the piece to a destination square before releasing it.

The project was developed iteratively, with particular attention to the mechanical and magnetic interface between the hidden actuator and the visible pieces. The main engineering challenge was not simply XY positioning, but achieving reliable magnetic coupling across the board while keeping the pieces stable, visually normal, and light enough to move without excessive friction. Tests showed that a 6 × 2 mm embedded magnet was sufficient for pawn prototypes, while heavier pieces required stronger coupling approaches such as 6 × 3 mm magnets and/or multiple magnets. The work also established practical CAD dimensions for the printed pieces and demonstrated the usefulness of a thin wooden top for maintaining a normal board appearance without sacrificing magnetic transfer.

This report documents the system requirements, architecture, mechanical and electrical design, control approach, development methodology, observed findings, limitations, and recommended next steps. Where quantitative measurements were not recorded during prototyping, the report deliberately presents qualitative findings rather than fabricated numerical performance data.

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1. Introduction

1.1 Motivation

The goal of Wizard Chess was to create the visual effect of chess pieces moving by themselves, inspired by fictional “wizard chess” while remaining grounded in a practical engineering implementation. The desired experience required the board to look and behave like an ordinary physical chessboard from the player’s perspective. Consequently, visible robotic arms, rails above the board, or motors inside individual pieces were undesirable. The actuation mechanism therefore had to be concealed below the playing surface.

This constraint naturally led to a magnetic transport concept: a movable magnetic field beneath the board can pull a magnetized chess piece across the top without any visible mechanical connection. The resulting problem is interdisciplinary, combining CAD, additive manufacturing, mechanical design, magnetics, stepper-motor motion, power conversion, and software-based coordinate control.

1.2 Project Objective

The primary objective was to build a functional prototype capable of automatically transporting selected chess pieces between board squares using a hidden XY mechanism. The design objectives were:

- Preserve a conventional chessboard appearance, with the actuation mechanism hidden underneath.
- Move pieces in two dimensions using a compact Cartesian gantry.
- Transfer force magnetically through the board surface.
- Use custom 3D-printed pieces that can contain permanent magnets without visibly altering their external form.
- Control motion using a GRBL-compatible command workflow and repeatable board coordinates.
- Develop the system iteratively through physical testing rather than relying only on theoretical assumptions about magnetic force and friction.

1.3 Scope

The project scope centers on physical piece transport. Full chess-engine integration, automated board-state sensing, capture management, and production-level reliability were not required for the initial prototype. The report therefore focuses on the mechanisms that were actually designed and tested: the board structure, XY drive, electromagnetic carriage, magnetized pieces, CAD/3D printing, power arrangement, motion-control concept, and observed behavior.

2. Engineering Requirements and Constraints

The design was governed by several requirements. Improving one parameter could worsen another; for example, a thicker board improves stiffness but weakens magnetic coupling, while a stronger permanent magnet can improve traction but may increase unwanted interactions with nearby pieces.

ID	Requirement	Engineering meaning	Prototype approach
R1	Concealed actuation	Motors, belts, and electromagnet should remain below the visible playing surface.	Core aesthetic requirement.
R2	Two-axis motion	Actuator must reach all relevant board coordinates.	Implemented using an XY gantry.

ID	Requirement	Engineering meaning	Prototype approach
R3	Magnetic transmission	Force must pass through the board top.	Board thickness and magnet selection are critical.
R4	Stable pieces	Pieces must stand normally and not visibly contain an actuator.	Magnets embedded in printed bases.
R5	Low friction	Piece drag must stay below available magnetic traction.	Affected by piece mass, base geometry, board finish, and magnet strength.
R6	Repeatable positioning	Source and destination squares must map to repeatable carriage coordinates.	GRBL-compatible motion control.

Central design trade-off: reliable motion depends on the complete force chain — electromagnet → board gap → embedded piece magnet → piece mass/friction — rather than on any one component in isolation.

3. System Architecture and Operating Principle

Wizard Chess can be interpreted as a hidden planar pick-and-place system in which magnetic attraction replaces a mechanical gripper. The actuator is constrained to an XY plane beneath the board. A permanent magnet inside each movable piece provides a magnetic target. The electromagnet is switched on only when coupling to a piece is required.

Wizard Chess System Architecture

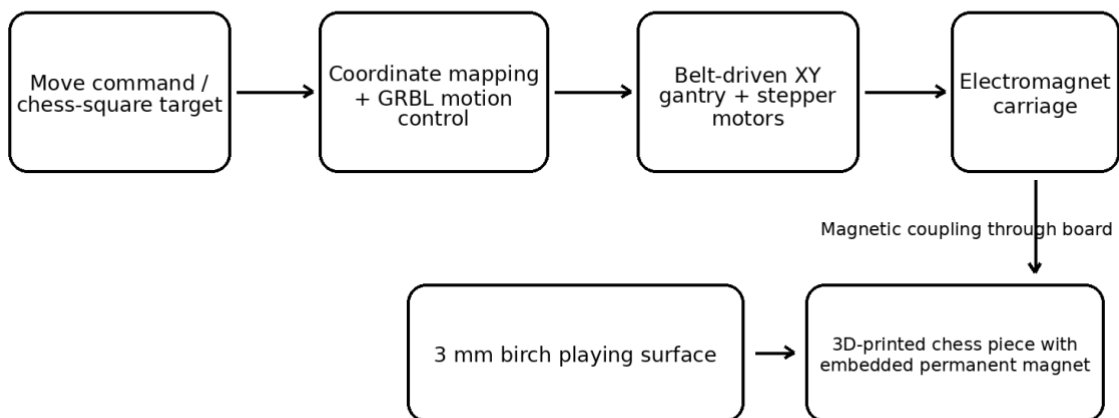


Figure 1. Functional architecture of the Wizard Chess prototype.

3.1 Operating Principle

1. A move is represented as a source square and a destination square.
2. Each square is mapped to a Cartesian position in the gantry coordinate system.
3. The carriage moves beneath the center of the source piece.
4. The electromagnet is energized so that it couples to the permanent magnet in the piece base.
5. The carriage translates toward the destination while the piece slides on the top surface.

6. At the destination, the electromagnet is switched off and the piece remains on the target square.

Automated Piece-Movement Sequence

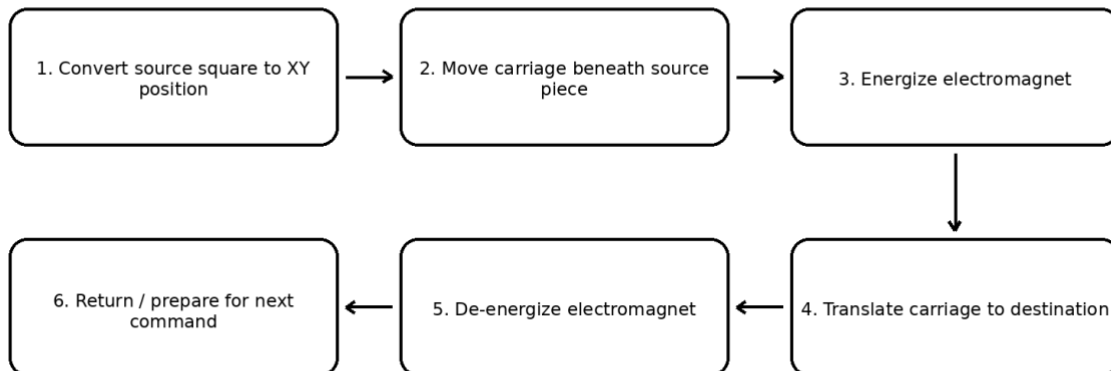


Figure 2. Representative automated movement sequence.

3.2 Coordinate Mapping

A chessboard is naturally discrete, but the gantry operates in continuous Cartesian coordinates. The software therefore requires a mapping from file/rank indices to physical XY positions. If the square pitch is denoted by s and the origin is defined relative to one board corner, a square-center mapping can be written in the general form $x = x_0 + (i + 0.5)s$ and $y = y_0 + (j + 0.5)s$, where i and j are zero-based file and rank indices. In practice, x_0 and y_0 are calibration offsets that account for the actual mechanical origin and the alignment between the visible board and hidden gantry.

The project used GRBL-style commands for motion control. This allows the carriage to be positioned using familiar G-code-like linear moves while leaving higher-level chess logic separate from the low-level motor controller.

4. Mechanical Design

4.1 Playing Surface and Board Envelope

The board top was designed around a 60 × 60 cm playing area with a 3 mm birch plywood surface. Birch plywood provides a useful compromise between rigidity, ease of fabrication, low thickness, and non-ferromagnetic behavior. The 3 mm thickness is especially important because magnetic force decreases strongly as the separation between magnets increases. The top therefore acts simultaneously as a visible game surface and as the principal air-gap component in the magnetic circuit.

Keeping the board visually ordinary also places a packaging constraint on the mechanism: the gantry, belts, motors, carriage, wiring, and electromagnet must fit below the surface without producing excessive overall height.

4.2 Linear Motion System

The hidden motion system uses a two-axis belt-driven gantry powered by two NEMA 17 bipolar stepper motors rated at 42 Ncm. Linear guidance is provided by five 8 mm × 500 mm rods together with LM8UU 8 mm linear bearings. The carriage is driven through GT2 timing belts using compatible 20-tooth motor pulleys and 20-tooth idler pulleys. This arrangement provides a compact planar motion system that can

be packaged below the playing surface while allowing the electromagnet to reach the required board coordinates.

Mechanically, the gantry must satisfy two different forms of accuracy. First, it must place the electromagnet sufficiently close to the centerline of the embedded piece magnet to establish strong coupling. Second, it must maintain alignment while the piece is dragged toward the destination without colliding with neighboring pieces. Belt tension, pulley alignment, LM8UU bearing alignment, rod parallelism, carriage stiffness, and secure M3 fasteners therefore directly influence magnetic reliability.

4.3 Custom 3D-Printed Chess Pieces

Custom chess pieces were produced by additive manufacturing so that permanent magnets could be integrated into the base. This was preferable to externally attaching magnets because it preserves appearance, protects the magnet, and makes the base geometry repeatable. The latest CAD design used a nominal base diameter of 23.63 mm for the documented pieces, with a magnet cavity of 6.10 mm diameter and 3.05 mm depth. The slight dimensional clearance around a nominal 6 mm magnet supports practical assembly despite printing tolerances.

Parameter	Value
Base diameter	23.63 mm
Rook height	65.14 mm
King height	65.50 mm
Knight height	65.25 mm
Magnet cavity diameter	6.10 mm
Magnet cavity depth	3.05 or 2.05 mm (Piece-Dependent)

Support strategy and printability were part of the iteration process. Organic and snug support approaches were considered during printing, with the practical aim of reducing support material and cleanup while preserving the detailed chess geometry.

4.4 Magnetic Coupling

Magnetic coupling proved to be the most sensitive mechanical interface in the project. The moving carriage carries a SEN-MAG 25 N electromagnet beneath the board, while the chess pieces contain 6 × 2 mm neodymium magnets in their bases. A pawn prototype could be moved successfully with this piece-magnet configuration. Heavier pieces were more difficult to move consistently because their greater mass and contact friction demanded more horizontal traction, leading to trials or consideration of thicker 6 × 3 mm magnets and/or multiple embedded magnets.

The result demonstrates why simple magnet “strength” specifications are insufficient by themselves. The usable pulling force at the piece depends on the spacing through the board, alignment between the two magnetic elements, the electromagnet field, the permanent magnet geometry, the coefficient of friction at the base, and the normal load from the piece. A configuration that works well for a pawn may not provide enough force margin for a king, rook, or other heavier model.

5. Electrical and Control Design

5.1 Hardware Components

The electrical and control system is centered on an Arduino Uno R3 fitted with an Arduino CNC Shield. Two DRV8825 stepper-driver modules control the X- and Y-axis NEMA 17 motors, while the SEN-MAG 25 N electromagnet provides the magnetic actuation used to engage the chess pieces. Jumper caps on the CNC Shield are used for driver configuration, and jumper wires provide the required signal interconnections.

Component	Quantity	Function
Arduino Uno R3	1	Main motion-control board for the GRBL-based workflow.
Arduino CNC Shield	1	Interfaces the Arduino with the stepper drivers and motor-power connections.
NEMA 17 bipolar stepper motor, 42 Ncm	2	Provides X- and Y-axis actuation.
DRV8825 stepper-motor driver	2	Drives the NEMA 17 motors from STEP/DIR control signals.
12 V, 3 A power supply	1	Main listed DC power source for the prototype.
SEN-MAG 25 N electromagnet	1	Carriage-mounted magnetic actuator used to couple to a chess piece.
Jumper caps and jumper wires	As required	Driver configuration and electrical interconnection.

5.2 Power Architecture

The prototype uses a 12 V, 3 A DC power supply as its main listed power source. The Arduino CNC Shield provides the interface for the motor-power rail and the installed DRV8825 modules, while the Arduino Uno supplies the logic and motion-control signals. Power distribution is therefore separated conceptually into the motor-power path and the controller/signal path, with the exact electromagnet switching path documented separately below.

5.3 Stepper Motors and Driver Connections

Two DRV8825 modules are installed on the CNC Shield, with one driver assigned to each motion axis. Each driver receives STEP and DIR control signals from the Arduino through the shield and drives one four-wire NEMA 17 bipolar stepper motor. Jumper caps beneath the driver positions can configure the selected microstepping mode.

5.4 Electromagnet Control

The SEN-MAG 25 N electromagnet is mounted to the moving carriage and is activated only when a chess piece must be picked up and transported. A typical move consists of positioning the carriage beneath the source square, energizing the electromagnet, translating to the destination square, and then de-energizing the electromagnet to release the piece.

5.5 Controller and GRBL Integration

The Arduino Uno R3 acts as the motion controller and uses a GRBL-compatible workflow. High-level software converts a chess move into source and destination board coordinates, while GRBL converts the resulting motion commands into timed STEP and DIR signals for the X- and Y-axis drivers. This separates chess-specific logic from low-level motor-pulse generation and makes calibration possible in physical Cartesian coordinates.

5.6 Electrical Connection Overview

Source	Destination	Purpose
12 V, 3 A DC power supply	CNC Shield motor-power input	Provides the main listed motor-power rail.
Arduino Uno R3	Arduino CNC Shield	Provides controller logic and GRBL motion signals.
CNC Shield X-axis driver position	DRV8825 → X-axis NEMA 17	Controls horizontal gantry motion.
CNC Shield Y-axis driver position	DRV8825 → Y-axis NEMA 17	Controls vertical gantry motion.
CNC Shield jumper positions	Jumper caps	Configures the selected DRV8825 microstepping mode.
Arduino control output	Electromagnet switching stage	Commands magnetic pickup/release; exact switch was not retained.
Switching stage	SEN-MAG 25 N electromagnet	Supplies switched current to the magnetic actuator.

6. Development Methodology

6.1 Iterative Prototyping

The project followed an iterative hardware-development methodology. Instead of fixing all dimensions and components before physical testing, subsystems were repeatedly built, evaluated, and modified. This approach was particularly appropriate for the magnetic interface because real-world performance depends on tolerances, material surfaces, piece weight, and field alignment that are difficult to predict precisely from nominal component specifications alone.

6.2 CAD and Additive Manufacturing

Chess-piece geometry was adjusted in CAD to create hidden magnet cavities while preserving the visual form and usable base. Magnet holes were revised to approximately 6.10 mm diameter and 3.05 mm or 2.05mm depth, and base/piece dimensions were refined across printing iterations.

A key advantage of additive manufacturing was rapid parameter iteration. Magnet depth, number of magnets, and base geometry could be changed without redesigning the complete board. This separated the “piece optimization” problem from the gantry design and made it possible to test stronger magnetic configurations for heavier pieces.

6.3 Subsystem-to-System Integration

The practical development sequence can be summarized as follows:

1. Validate that a magnetized test piece can be attracted through the chosen top-board material.
2. Build and verify two-axis carriage motion independently of the chess pieces.
3. Mount the electromagnet to the moving carriage and route power so that it can remain active during translation.
4. Print pieces with embedded-magnet cavities and compare different magnet dimensions/configurations.
5. Calibrate board-square positions against the gantry coordinate system.

6. Perform integrated moves and observe loss of coupling, unwanted piece rotation, collisions, or insufficient traction.
7. Revise magnet configuration, piece geometry, or mechanical settings based on observed failures.

7. Testing and Findings

7.1 Magnet Size Versus Piece Load

The clearest experimental finding was that one magnet configuration did not perform equally across all chess pieces. A 6 × 2 mm magnet was adequate for the pawn prototype, but heavier pieces were more likely to resist motion or lose reliable coupling. Increasing magnet thickness to 6 × 3 mm and/or using two magnets was explored as a way to raise the available magnetic interaction.

Finding 1 — Piece-specific force margin: magnet selection should be based on the heaviest/highest-friction piece, or pieces should be designed to have more uniform mass and base friction.

7.2 Board Thickness Is a First-Order Variable

The use of a 3 mm birch top was not merely a cosmetic decision. Every additional millimeter of separation weakens magnetic interaction, so the playing surface has a direct effect on transport reliability. At the same time, the top must remain sufficiently stiff and flat to avoid local deflection or inconsistent spacing above the gantry.

Finding 2 — Thin structural surface: the board top should be treated as part of the actuation system, not as a passive enclosure panel.

7.3 Alignment Matters as Much as Nominal Magnet Strength

When the electromagnet is centered below the permanent magnet, the coupling is strongest and the available horizontal pulling force is most useful. Offset alignment reduces force and can encourage the piece to rotate or detach. Therefore, gantry calibration and repeatable square-center mapping directly affect magnetic reliability.

Finding 3 — Coupled calibration problem: motion calibration and magnetic performance cannot be tuned independently; coordinate error effectively becomes a loss of magnetic force.

7.4 Friction and Piece Geometry

The project exposed friction as a practical limiting factor. A piece can remain strongly attracted vertically yet still fail to slide if the horizontal component of the coupling force is insufficient. Taller and heavier pieces also have greater inertia and may be more prone to tipping or rotation during aggressive acceleration. This supports using controlled acceleration and a low-friction, flat base surface rather than attempting to solve every failure by increasing magnet strength.

7.5 Integrated Prototype Result

The prototype reached the stage of automatically transporting chess pieces using the concealed magnetic carriage concept, validating the fundamental architecture. The development process also revealed that reliable operation across a full chess set requires systematic tuning rather than a single successful proof-of-concept move. The most important result is therefore both functional and methodological: the architecture works, but robustness depends on designing the board, magnets, piece masses, motion profile, and calibration as one coupled system.

8. Engineering Challenges and Design Decisions

Challenge	Design response	Important consequence
Magnetic force too low for heavier pieces	Increase permanent-magnet thickness/count; reduce friction; verify electromagnet alignment.	Stronger magnets improve traction but can increase attraction to unintended nearby pieces.
Board must look normal	Use thin birch top and hide complete mechanism below it.	Thin top benefits magnetics but requires adequate support for stiffness.
Pieces must retain normal appearance	Embed magnets inside 3D-printed bases.	Requires CAD tolerance and controlled cavity depth.
Accurate pickup needed	Calibrate square centers in XY coordinate system.	Mechanical backlash/belt tension can appear as magnetic unreliability.
Different pieces have different mass/friction	Iterate magnet configuration and geometry.	A uniform internal ballast/magnet standard could simplify production.

9. Prototype Assessment

Against the original project objectives, the prototype can be assessed as follows:

Objective	Assessment	Evidence / comment
Hidden mechanism	Achieved in concept/prototype	XY gantry and magnetic carriage located below board surface.
Automatic physical movement	Demonstrated	Core magnetic transport concept validated.
Conventional board appearance	Achieved by design	3 mm wooden top separates visible board from mechanism.
Custom magnetized pieces	Achieved	3D-printed bases incorporate magnet cavities.
Uniform reliability across all pieces	Partially achieved	Heavier pieces required stronger coupling iterations.

The project should therefore be considered a successful mechatronic proof of concept with an identified path toward a more repeatable second-generation system. Its strongest contribution is the integration of hidden planar motion with custom magnetized pieces in a form factor that preserves the tactile and visual qualities of a physical chessboard.

10. Limitations and Risk Analysis

10.1 Current Limitations

- No retained quantitative data for positioning error, maximum speed, pull force, or move success rate.
- Different piece types do not yet have demonstrated equal coupling margin.
- Board-state sensing is not part of the current reported prototype.
- Long-duration reliability, belt stretch, thermal behavior of the electromagnet, and repeated-cycle wear have not been characterized.

10.2 Technical Risks

Risk	Effect	Mitigation
Loss of magnetic coupling during a move	Piece stops between squares	Increase force margin, reduce acceleration, improve alignment, reduce friction.
Adjacent piece disturbance	Incorrect board state	Control path, field strength, activation duration, and magnet orientation.
Carriage coordinate drift	Pickup/release off-center	Homing, periodic calibration, rigid belt/pulley mounting.
Electromagnet overheating	Reduced reliability or component damage	Characterize current/duty cycle and add thermal limits if required.
Piece tipping/rotation	Unreliable placement	Lower center of mass, smooth acceleration, optimized base.
Power switching transients	Controller instability/component stress	Use suitable transistor/MOSFET driver and flyback protection.

11. Future Development

A future version of the Wizard Chess system could integrate magnetic sensors underneath the chessboard to detect the position of each piece in real time.

This would allow one player to move the chess pieces by hand, just like on a normal chessboard. The sensors would register the movement and update the digital board state automatically. The move could then be sent to the chess engine.

After the AI selects its response, the hidden XY gantry and electromagnet system could automatically move the corresponding piece on the physical board. This would create a more natural human-versus-AI experience, where the human player interacts directly with the board while the AI responds physically without requiring any manual input through a computer or interface.

One possible extension would be to connect the system to Lichess, allowing the physical board to be used for games against real players over the internet. A move made manually on the board could be detected by the

sensors and transmitted to the online game. When the remote player responds, the hidden XY gantry and electromagnet system could automatically reproduce that move on the physical board.

This would transform the project from a remotely controlled prototype into a more complete interactive chess system capable of supporting both human-versus-AI and human-versus-human online gameplay through a physical chessboard.

12. Conclusion

Wizard Chess demonstrates that a conventional-looking chessboard can be given autonomous physical movement using a concealed XY gantry and magnetic coupling. The prototype combines a 40 × 40 cm board, a thin 3 mm birch playing surface, belt-driven stepper motion, a carriage-mounted electromagnet, and 3D-printed chess pieces containing permanent magnets. GRBL-compatible positioning provides a practical interface between chess-square coordinates and the hidden Cartesian mechanism.

The most important engineering lesson from the project is that successful magnetic transport is a system-level problem. The board thickness, magnet separation, piece mass, base friction, actuator alignment, speed, acceleration, and gantry calibration all influence whether a move succeeds. Testing showed this directly: the 6 × 2 mm magnet configuration was sufficient for pawns, while heavier pieces required stronger magnetic configurations and further iteration.

The project achieved its core proof-of-concept objective and established a strong foundation for a second-generation design. Future work should focus on quantitative characterization, standardized piece coupling, safe and documented electromagnet drive electronics, collision-aware path planning, board-state sensing, and long-duration reliability testing. With these improvements, the concept can progress from an impressive autonomous demonstration to a repeatable and fully interactive automated chess system.

Appendix A. Known Prototype Dimensions and Parameters

Item	Documented Parameters
Playing surface	40 × 40 cm
Top material	Birch plywood
Top thickness	3 mm
Motion architecture	2-axis GT2 belt-driven XY gantry
Actuator	SEN-MAG 25 N electromagnet
Main power supply	12 V, 3 A DC
Control interface	GRBL-compatible motion commands
Piece manufacture	FDM 3D printing
Piece base diameter	23.63 mm
Rook height	65.14 mm
King height	65.50 mm

Item	Documented Parameters
Knight height	65.25 mm
Magnet cavity	Ø 6.10 mm × (2.05 or 3.05 mm, piece-dependent)
Pawn magnet trial	6 × 2 mm
Heavier-piece trials	6 × 3 mm and/or dual magnets

Appendix B. Representative Motion-Control Sequence

The following sequence illustrates the control logic without claiming the exact commands or machine coordinates used in the final prototype. Coordinates X_s/Y_s and X_d/Y_d represent calibrated square centers.

- 1. Home / establish coordinate reference*
 - 2. Rapid or controlled move to (X_s, Y_s)*
 - 3. Electromagnet ON*
 - 4. Linear XY move to (X_d, Y_d) using a conservative feed/acceleration profile*
 - 5. Electromagnet OFF*
 - 6. Move carriage to an idle position or begin the next command*
-

A production implementation should add homing checks, occupancy/path validation, electromagnet timing, error handling, and state confirmation around this basic sequence.